

Assignment-1

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① What is IOT ? Explain in simple words with one example:

→ The Internet of Things (IoT) means connecting everyday physical objects to the internet, allowing them to communicate with each other, share data, and perform tasks automatically. For ex: a "smart" air conditioner (AC) can sense that a room is too hot and turn itself on automatically, even when you're at home.

② Write any five features of IoT devices:

→ **Connectivity:** They must be able to connect to the internet or other devices to send and receive data.

Sensors: They have sensors to gather info from their surroundings, like temperature or motion.

Processing Power: They contain a small chip that acts like a brain to understand data and make decisions.

Take Action: Based on the data, the device can automatically perform a task, like switching on a light.

Security: Have security features to protect the data.

③ What is the use of sensors in IoT?

→ In IoT, sensors are like "eyes and ears" of a device. Their primary use is to collect data and information from the physical environment, such as temperature, light, motion, or air quality. It can be used in:

- Smart homes
- Health & fitness
- Transportation
- Agriculture
- Industry

④ Differentiate between embedded devices and IoT devices?

IOT devices	Embedded devices
Is a small device that can connect to internet and share data.	A device that do a fixed job using a small computer inside.
It can connect to internet	It cannot connect to internet.
Smart control through the help of mobile phone	Does not support any smart control
can get updates online	Cannot get's updates
Ex: smartwatch, smart city, motion sensors with data detection / collection	Eg: washing machines, digital watches, calculators, microwave ovens.

⑤ Name five devices that are IoT devices and explain.

→ Smart Bulbs: These light bulbs connect your phone via, Wi-Fi, allowing you to turn them on or off.

Smart watch: Monitors your heart rate and steps and gives suggestions on Phone app.

Smart Door locks: Automatically locks the door from anywhere using a mobile device

Smart Dust bin: Sends alert message when they are full.

Soil moisture sensor: An agricultural IoT devices that checks the soil and tells a farmer when it's time to water the plants.

⑥ Explain what is meant by 'smart' in smart devices.

→ In the context of IoT, smart means a device can operate intelligently and automatically without constant human control. A device is 'smart' because it can:

- sense its environment (e.g. know the temperature is hot).
- process that information to make a decision (e.g. decide it needs to cool the room)
- Act on that decision automatically (e.g. turn the AC on by itself)

⑦ What is the use of internet in IoT?

→ The internet is essential in IoT for connectivity and data sharing. It allows devices to send the data they collect from sensors to a cloud server or a mobile app. It also allows users to receive updates and send commands back to the device to control it from a distance.

⑧ List 3 types of IoT boards and describe in brief.

i) Arduino uno

- A popular board for beginners that is used for learning how to control sensors, LEDs, and motors in basic electronic and home automation projects.

ii) ESP32

- A low-cost microcontroller that has built-in Wi-Fi and Bluetooth, making it ideal for wireless IoT projects like smart home devices.

iii) Raspberry Pi 4 model B

- A powerful mini-computer that can run a full OS like Linux.

⑨ What is IoT ecosystem? List its parts.

→ An IoT ecosystem is the complete system of all components needed for IoT devices to work together.
Parts are?

- Device / Sensors: collect data from the real world.
- Network: sends the data using technologies like Wi-Fi or 4G.
- Data processing: converts raw data into useful info.
- Cloud / storage: stores the data safely online.
- Application / APP: the interface that the user interacts with.
- User: The person who uses the IoT system.

⑩ Write 5 examples of IoT in daily life.

- i) Turning on your home lights from your office using your phone.
- ii) A fan that turns on automatically when the room gets hot.
- iii) A smartwatch that tracks your heart rate and sleep.
- iv) Smart traffic lights that adjust their timing to reduce traffic jams.
- v) Getting a real-time alert on your phone if there is a gas leak at home.

(11) Explain how IOT helps in home automation.

→ IOT helps in home automation by connecting home appliances to the internet, allowing them to be controlled remotely and work automatically.

Example:

You can use your phone to turn on lights or lock your door from anywhere. Devices can also act on their own, such as light turning off when no one is in the room to save energy or an AC turning on when it gets hot.

(12) List IOT applications in agriculture.

→ IOT applications in agriculture help make farming easier and more efficient. Key applications include:

- Soil moisture sensors to tell farmers exactly when to water their plants.
- Smart irrigation systems that can water fields automatically based on sensor data.
- Weather sensors that provides farmers with real-time climate information.

(13) What is an actuator? How is it used in IOT?

→ An actuator is a component in the IOT system that is responsible for taking a physical action. After a sensor collects data and the system processes it to make a decision, the actuator is the part that carries out the action. For example, in a smart AC system, the processor decides it's too hot, and the actuator is the mechanism that actually turns the AC's cooling unit on. Other examples include an actuator unlocking a door or starting a water pump.

(14) Name and describe any 2 layers of IOT architecture.

→ 1) Sensing Layer (Device Layer)

- This is the first layer and its job is data gathering. It consists of the physical objects, primarily sensors and actuators, that collect data from the environment.

2) Application Layer

- This is the top layer of the architecture that the end-user interacts with. It provides user-friendly interfaces like mobile apps or web portals that allow a person to control the IOT devices and view the data that has been collected and analyzed.

(15) What are the components of IOT solution architecture?

→ The IOT solution architecture, or ecosystem, consists of several key components that work together:

- **Device / sensor**: The hardware that gathers data from the environment.
- **Network**: The communication technology (wifi, bluetooth) that transmits the data.
- **Data Processing**: The system that analyzes data to make it.
- **cloud / storage**: A safe online repository for storing all the collected data.
- **Application / APP**: The user interface for monitoring and controlling the system.

(16) Explain IOT architecture in detail with example.

→ The IOT architecture is a 4-layer structure that shows how data moves from collection to the user.

- **Sensing Layer**: The base layer with sensors and actuators that gathers data from the environment.

- Network Layer
 - Transmits the collected data using networks like wi-fi, Bluetooth, or 4G/5G.
- Data Processing Layer:
 - This layer analyzes the raw data to extract useful information and make decisions.
- Application Layer
 - The top layer that the user interacts with through an app or web portal to manage the system.

Example

- A smart AC's sensor (sensing) detects it's hot, sends this data over wi-fi (Network). a server decides to cool the room, and you can control it from your phone app.

(17) Describe the working of IoT with a neat block diagram.

→ IoT devices work together from sensing the data from the environment, collecting it to the cloud through the internet and performing basic operations on the application part, and gives result to the user.

1) sensor layer

- Also called perception layer, it sense the data from the environment like temperature, humidity. Devices used are sensors, actuators, identification.

(i) Network Layer

- used to transfer data from sensing devices to the cloud using internet or communication networks.

technology: wifi, ethernet, 4G, 5G

Protocol: HTTPs, IPV4, IPV6, TCP, IP, UDP

(ii) Data processing Layer

→ This layer stores the data, processes and analyze the data. Also makes decisions.

Tech stack: Cloud Platform, Database, API, ML, etc.

(iii) Application Layer

→ This layer displays the info to the user and allows them to control the IoT devices.

ex: mobile applications, web applications.

Application Layer



Data processing layer



Network Layer



sensor Layer

⑧ Compare embedded system and IOT devices with 5 points.

→ Here is a five-point comparison between Embedded and IOT devices:

Feature	Embedded Devices	IOT Devices
i) Internet	Does not connect	connects to the internet
ii) Data sharing	Does not share data.	can send and receive data online
iii) work type	works alone, fixed function.	works with the internet or other devices.
iv) Smart features	No smart control.	smart control via phone or voice.
v) updates	cannot be updated	can get updates via the internet.

⑨ write a detailed note on IOT solution architecture.

→ An IOT solution architecture is the structured design of how an IOT system works, from data collection to user interaction. It is composed of four layers through which data flows from bottom to top.

- Sensing Layer: Gathers data
- Network Layer: Transmits data
- Data processing Layer: Analyze data and makes decisions.
- Application Layer: Provides a user interface like a mobile app.

②② What is an IoT ecosystem? Describe all its components.

→ An IoT ecosystem is the complete system needed for IoT devices to work together, including hardware, software, and networks.

Its components are:

- Devices / sensor: collects data from the world.
- Network: sends data (e.g. wi-fi, 4G).
- Data processing: converts data into useful info.
- Cloud / storage: stores the data safely online.
- Application / App: Allows the users to see and control the devices.
- User: The person who uses the system.

②③ Describe any five IoT boards and their usage.

→ The notes details four major IoT boards.

- Arduino uno: A beginner-friendly board used for learning electronics and making small robots.
- ESP32: A board with built-in wifi and bluetooth, perfect for wireless IoT projects and smart home devices.
- NodeMCU (ESP8266): A very low-cost wifi board used for simple home automation and sending sensor data online.
- Raspberry Pi 4 model B: A powerful mini-computer for advanced projects like AI, robotics, or running a home automation server.

Q22 Explain the role of sensors and actuators in IoT.

- Sensors: Their role is to be the "eyes and ears" of an IoT device, collecting data like temperature, light or motion from the physical world.
- Actuators: Their role is to perform a physical action based on the data, like turning on a motor, unlocking a door, or switching on a light.

Q23 Discuss challenges and security issues in IoT.

→ The provided notes ~~do not~~ discuss challenges or security issues like not a major issue as the data is already encrypted before transferring.

Security is a key feature of IoT devices, designed to protect data from being stolen or misused through features like passwords, encryption and authentication.

Q24 Explain the decision-making framework used in IoT.

→ The IoT Decision Framework is a step-by-step guide to help plan and build IoT systems by making smart choices. It focuses on six key areas to think about.

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|--------------------|---------------|
| i) User experience | iv) Security |
| ii) Data | v) Operations |
| iii) Business | vi) Devices |

Q3) Discuss the applications of IOT in various domains.

→ IOT has applications across many domains:

- Smart home: Remotely controlled lights and door locks.
- Wearable: Smartwatches and fitness bands that track health metrics like heart rate and steps.
- Healthcare: Smart heart monitors and glucose monitors that send data to doctors.
- Smart cities: Smart traffic lights to manage jams and smart dustbins that signal when they are full.
- Smart Agriculture: Soil moisture sensors and smart irrigation systems to save water and improve crops.
- Smart vehicles: Cars with GPS for navigation and self-driving cars.